

IIEViT-ROI: An Efficient ROI-Guided CNN–Transformer for Explainable Chest X-ray Disease Classification

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Abstract. Subtle and overlapping radiographic patterns make chest X-ray (CXR) triage challenging, particularly in low-resource settings with limited access to radiologists. We propose Improved Input Enhanced Vision Transformer with Region of Interest (IIEViT-ROI), an explainable and computationally efficient hybrid CNN–Transformer framework for multi-class lung disease classification from CXRs. The model combines lung region-of-interest (ROI) segmentation, attention-augmented VGG19 feature extraction, ROI-gated patch embedding, CNN–Transformer feature fusion, and class-token attention visualisation to focus prediction on anatomically relevant image regions while preserving global context. We evaluate IIEViT-ROI on three public CXR datasets–Lung Disease, COVID-19 Radiography, and Covid-n-Pneumonia and assess cross-domain adaptation using a small anonymised clinical cohort collected from six healthcare facilities. Compared with X-Vision, IIEViT-ROI reduces inference latency from 6.70–7.75 ms to 0.88–1.16 ms in our test environment, achieving approximately 6–8 $\times$ faster inference with fewer parameters (41.02M vs. 74.1M). Although it incurs a modest 1–2% accuracy drop on some benchmark datasets, it achieves stronger transfer-learning performance on the clinical cohort, indicating improved adaptability under domain shift. These results suggest that IIEViT-ROI offers a promising accuracy–efficiency trade-off for retrospective CXR screening research, while prospective clinical validation remains necessary before deployment.

Keywords: ROI · Explainable AI · Lung Disease · Deep Transfer Learning · Chest X-ray

1 Introduction & Background Study

Medical imaging has assumed central importance amid the rapid emergence and global spread of respiratory diseases such as COVID-19, demanding diagnostic tools that are accurate, scalable, and timely [18, 21]. Among available modalities, chest X-ray (CXR) remains widely deployed for detecting pulmonary abnormalities due to its non-invasiveness, low cost, and rapid acquisition [14, 4, 24]. However, conventional radiological interpretation is time-consuming and subject to inter-reader variability, challenges that are exacerbated in low--resource settings with limited access to expert radiologists [25, 23, 19].

These pressures are acute in low and middle income countries, where the physician-to-population ratio remains critically low (such as, a South Asian country, Bangladesh, has approximately 0.48 doctors per 1,000 people and for Uganda the ratio is 0.13 per 1,000 according to WHO, 2024). Limited availability of advanced imaging further restricts diagnostic options, motivating automated and dependable systems that neither increase clinician workload nor compromise interpretability and clinical trust [1, 16].

Deep learning models based on CNNs and Vision Transformers (ViTs) have shown strong performance for CXR analysis [22, 17, 28], but clinical translation remains hindered by (i) dataset shift and limited cross-domain generalisation, (ii) computational cost that impedes high-throughput screening, and (iii) limited transparency that reduces clinical confidence [25, 12, 4]. Prior CXR systems illustrate these trade-offs: Alshmrani et al. [1] noted interpretability limitations in multi-class CXR classification; IEViT [25] reported cross-dataset gains with increased computational complexity; PulDi-COVID [6] achieved strong performance but required 8.59s per CXR; and X-Vision [7] combined XceptionNet and a ViT via dynamic fusion but reported restricted dataset diversity, limited explanatory capability, and untested robustness to noise. CheXNet [26] performed radiologist-level detection of pneumonia on 14 diseases using DenseNet-121 (8M parameters) but does not leverage post-hoc Class Activation Mapping to identify anatomy nor tested in cross-domain settings suitable for deployment in resource-limited settings. SwinCheX [29] applies the hierarchy of shifted windows of the Swin Transformer to multi-class CXRs classification, obtaining a mean AUC of 0.810 on ChestX-ray14; however, it comes at a high computational cost with 197M parameters and only gives post hoc Grad-CAM results, while ignoring anatomical priors. The BioViL [8] and BioViL-T [5] architectures involve using a ResNet-50-based image encoder together with a language model tuned specifically for medical imaging in a vision-language setting, providing impressive phrase grounding performance, although such models cannot be employed without the corresponding text report that does not exist in low-resource settings.

While existing methods such as X-Vision [7] (99% on three classes, ~ 74.1 M parameters, 6.70–7.75 ms inference) achieve high in-domain accuracy, our cross-dataset analysis showed substantial performance deterioration on variable class cohorts, indicating overfitting and weak adaptability to unseen distributions. In this paper, we propose a hybrid CNN-Transformer framework, Improved Input Enhanced Vision Transformer with Region of Interest (IIEViT-ROI), for multi-class CXR classification that combines a VGG19 backbone [15] for robust local feature extraction [1] with Transformer reasoning for global context [25]. Unlike X-Vision’s parallel fusion, we incorporate lung region-of-interest (ROI) segmentation [27] to focus inference on anatomically relevant structures and suppress spurious activations from non-lung background regions, and we integrate explainability to produce visual attributions that support clinical interpretation [12, 19]. While saliency methods can be ambiguous or noisy [2], our ROI-guided design seeks more reliable, clinically meaningful explanations.

Hybrid CNN–Transformer architectures have been widely explored in medical imaging. TransUNet [10] and Swin-UNet [9] use transformer encoders for pixel-level segmentation tasks and are not designed for multi-class classification or real-time inference on edge hardware. IEViT [25] applies a ViT with cross-dataset training for CXR classification but lacks explicit lung ROI guidance during tokenisation and reports inference latency of 44.15–59.00 ms — prohibitive for high-throughput screening. X-Vision [7] achieves high in-domain accuracy via parallel XceptionNet–ViT fusion but uses no anatomical priors, provides limited explanatory capability, and its 74.1M parameter footprint and 6.70–7.75 ms latency restrict deployment on resource-constrained hardware. IIEViT-ROI addresses these gaps through two specific design choices absent in prior work: (i) ROI-gated adaptive patch embedding: lung ROI masks produced by the segmentation module are fused directly into the ViT tokenisation stage via a learnable confidence-aware gating function, so anatomical priors influence feature extraction from the earliest representation stage rather than being applied post-hoc; (ii) Alternating cross-attention fusion: CNN-derived global features interact with ViT patch tokens at every other transformer layer, enabling persistent multi-scale reasoning rather than a single fusion step. Together, these mechanisms fill the gap of combining anatomically-guided tokenisation with efficient, deployment-ready multi-scale reasoning — a combination not present in TransUNet, Swin-UNet, IEViT, or X-Vision.

In resource-constrained environments, deployment feasibility often outweighs marginal accuracy gains. On certain benchmarks, X-Vision performs 1–2% better than our model; nevertheless, its parameter count and inference time limit scaling for high-throughput screening. With 41.02M parameters and 0.88–1.16 ms latency, our model achieves substantially lower per-image latency than X-Vision under identical batch-size-16 evaluation conditions, as reported in Table 1. Our results show computational tractability for real-time inference, which is a prerequisite to time-sensitive screening situations [30].

To address dataset shift and improve relevance to underrepresented populations, we initiated real-world CXR data collection across six clinical facilities, complementing public datasets whose demographics and care pathways are primarily drawn from high-resource settings. Cross-domain evaluation demonstrates strong transfer learning capability: clinical fine-tuning achieved 95.56% validation accuracy vs. X-Vision’s 88.89%, validating adaptability with limited labeled data.

Key contributions of our work include:

1. *IIEViT-ROI*: a ROI-gated adaptive patch embedding mechanism and cross-attention CNN–ViT fusion module for multi-class CXR classification with integrated explainability. These two mechanisms – the confidence-aware gating of anatomical priors into ViT tokenisation, and the alternating cross-attention between CNN global features and ViT patch tokens – constitute the architecturally novel components of this work. The U-Net segmentation backbone [27], VGG19 feature extractor [1], and standard ViT encoder [25] are established components integrated within this novel fusion framework.

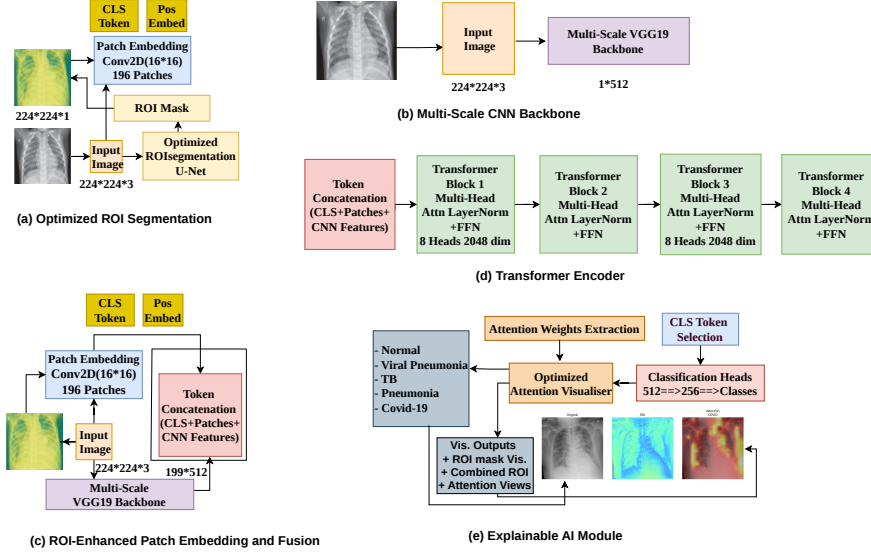


Fig. 1: Architecture of the proposed IIEViT-ROI model

2. A deployment-oriented design achieving $6\text{--}8\times$ faster inference than X-Vision [7] (0.88–1.16 ms vs. 6.70–7.75 ms), enabled by reduced sequential convolution bottlenecks and patch-parallel ViT computation (see Section 3.1 for full inference setup).
3. A multi-institutional CXR cohort with expert-validated annotations, used for transfer-learning evaluation under domain shift. Cross-domain results show improved generalisation over X-Vision (96.36% vs. 90.90%), indicating better adaptability with limited labelled data.

2 Methodology

This study proposes the IIEViT-ROI, a hybrid deep learning model that integrates spatial, contextual, and multi-scale features to classify lung diseases from CXRs with high accuracy and clinical interpretability. By leveraging ROI-based spatial priors and multi-scale convolutional features, it aligns automated predictions with radiologist-driven diagnostic reasoning.

2.1 Proposed IIEViT-ROI Model Overview

The IIEViT-ROI architecture (Fig. 1) integrates ROI-based segmentation, multi-scale CNN feature extraction, vision transformers with cross-attention, and explainable AI (XAI) to enable accurate and automated lung disease classification from chest radiographs. By capturing pixel-level morphology, structural

patterns, and contextual interactions, the system ensures comprehensive feature representation.

The architecture consists of five interconnected components: (1) Optimised ROI Segmentation, (2) Multi-Scale CNN Backbone with Channel Attention, (3) ROI-Enhanced Patch Embedding and Fusion, (4) Cross-Attention and Transformer Encoder, and (5) XAI Module collectively forming a progressively enriched representation for reliable disease characterization.

Optimised ROI Segmentation module (Fig. 1 (a)) uses a lightweight pre-trained supervised U-Net to isolate anatomically relevant lung regions to suppress ribs and background artifacts. The encoder expands channels ($3 \rightarrow 32 \rightarrow 64 \rightarrow 128$) using 3×3 convolutions and pooling, while the decoder restores spatial resolution via transpose convolutions with skip connections. A spatial attention mechanism refines segmentation:

$$A_{\text{spatial}} = \sigma(\text{Conv}_{7 \times 7}(F_{\text{dec}})). \quad (1)$$

The resulting ROI mask guides feature learning through weighted enhancement:

$$x_{\text{final}} = 0.7x + 0.3x_{\text{enhanced}}, \quad (2)$$

ensuring emphasis on clinically significant lung structures without suppressing contextual cues.

Multi-Scale CNN Backbone, a modified VGG19 backbone (Fig. 1(b)), extracts features from the CXR image at three processing scales (56×56 , 28×28 , 7×7). Early 10 layers are frozen to preserve low-level representations and make the computation efficient, while deeper layers are fine-tuned for medical adaptation. Channel attention enhances discriminative features, and scale-wise features are projected and adaptively fused into a unified 512-dimensional representation for transformer compatibility.

Specifically, features are extracted at three VGG19 pooling stages: **pool-2** (layers 1–8, output 56×56 , 128 channels), **pool-3** (layers 9–17, output 28×28 , 256 channels), and **pool-5** (layers 18–36, output 7×7 , 512 channels). Each scale is projected to a 512-dimensional space via a 1×1 convolutional projection layer, then fused using adaptive softmax weights $w \in \mathbb{R}^3$ learned during training.

ROI-Enhanced Patch Embedding and Fusion: In this module (Fig. 1 (c)) images are divided into 196 (14×14 grid) non-overlapping patches using depthwise convolution (stride 16). ROI information is processed in parallel and adaptively fused with patch embeddings through a gating mechanism, allowing confidence-aware integration of anatomical priors. Global CNN features and transformer class tokens are concatenated and projected as per the following equation:

$$F_{\text{fused}} = \text{Linear}(\text{Concat}(T_{\text{cls}}, F_{\text{CNN}})), \quad (3)$$

It forms a unified representation that captures both global context and local abnormalities. The resulting transformer input sequence consists of 199 tokens, including 196 patch tokens generated from the 14×14 patch grid, one learnable CLS token, and two CNN-derived global feature tokens. Each token is represented in a 512-dimensional embedding space, resulting in an input tensor of size 199×512 .

The gating function G operates as a continuous soft gate in $[0, 1]$ via sigmoid activation; no discrete confidence threshold is applied. The model learns the appropriate gating balance end-to-end during training, allowing confident ROI regions to amplify patch embeddings while uncertain regions default to the original patch representation.

Cross-Attention enables interaction between CNN-driven global features and ViT patch tokens, facilitating multi-scale reasoning. The **Transformer Encoder** (Fig. 1(d)) employs multi-head attention (8 heads per block across 4 Transformer encoder blocks, with an FFN dimension of 2048 — consistent with Fig. 1(d); the range “4–6 heads” refers to the hyperparameter search space, and the final deployed configuration uses 8 heads throughout) with learnable residual scaling:

$$x_{\text{out}} = x + \alpha \cdot \text{Module}(\text{LayerNorm}(x)), \quad (4)$$

improving gradient flow and training stability while maintaining computational efficiency.

In **Explainability** module (Fig. 1(e)), the attention weights from the class token to patch tokens are aggregated and reshaped into spatial maps, then overlaid with ROI masks to generate clinically interpretable visualizations highlighting diagnostically influential regions.

2.2 Training and Optimisation

To address class imbalance, we combine weighted cross-entropy and focal loss:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{CE}}^{\text{weighted}} + \lambda_2 \mathcal{L}_{\text{focal}}. \quad (5)$$

In our implementation, $\lambda_1 = 0.6$ and $\lambda_2 = 0.2$, with an auxiliary classification loss weighted at 0.2, giving the full objective:

$$\mathcal{L} = 0.6 \mathcal{L}_{\text{CE}}^{\text{weighted}} + 0.2 \mathcal{L}_{\text{focal}} + 0.2 \mathcal{L}_{\text{aux}}, \quad (6)$$

where the focal term $\mathcal{L}_{\text{focal}} = -\mathbb{E}[(1 - \hat{p}_y)^2 \log \hat{p}_y]$ emphasises hard-to-classify minority classes such as tuberculosis and bacterial pneumonia, and label smoothing ($\epsilon = 0.1$) is applied within $\mathcal{L}_{\text{CE}}^{\text{weighted}}$.

Optimisation uses AdamW optimiser, a decoupled weight decay variant of Adam, with differential learning rates for segmentation and backbone modules,

cosine annealing scheduling, gradient clipping, and strong data augmentation.

Transfer Learning: For domain adaptation on independent clinical datasets, pretrained weights are initialized, early layers are frozen, and the classification head is reconfigured. Fine-tuning with reduced learning rates and balanced sampling enables adaptation to domain shifts while preserving generalisable representations.

The ROI-weighted image enhancement (Eq. 2) is motivated by the well-established finding that focusing feature extraction on anatomically relevant regions reduces background noise and improves sensitivity for subtle pathological patterns [19]. The learnable gating in the patch embedding (Section 2.1) is preferred over hard masking because it allows the model to down-weight rather than discard non-lung context, preserving cues such as pleural effusion boundaries that extend beyond the strict lung boundary. The alternating cross-attention schedule (Fig. 1(d): blocks 1, 3 use cross-attention; Fig. 1(d): blocks 2, 4 in use self-attention only) balances the computational overhead of cross-attention against the benefit of repeated CNN-ViT interaction: ablation results show that full cross-attention at every block does not improve accuracy but increases inference time by approximately 0.18 ms. The combined loss function (Eq. 5) uses label smoothing ($\epsilon = 0.1$) to prevent overconfident predictions on the imbalanced training set, class-weighted cross-entropy to correct for frequency imbalance, and a focal term to further emphasise hard examples — a combination empirically validated in class-imbalanced medical imaging tasks. The ablation study in Section 4 provides empirical evidence for the contribution of each component.

This integrated design balances accuracy, interpretability, and computational efficiency, making the proposed IIEViT-ROI model, a candidate for future clinical validation in resource-constrained settings.

3 Experimental Results and Analysis

This section presents a comprehensive experimental evaluation of the proposed IIEViT-ROI model and comparative analysis with existing approaches.

To ensure a fair comparison and reliable ground truth, we conducted an extensive search across public repositories (e.g., GitHub) and found that only the VGG19+CNN [1] implementation was available. Hence, we implemented the existing methods ourselves, following their original descriptions and evaluated them on the same datasets. We attempted to validate our reimplementations against the originally reported results. For X-Vision, the original implementation was not publicly available and the exact training dataset split was not specified in the paper; our reimplementation achieved 96.53% on the three-class Covid-n-Pneumonia Dataset under our unified preprocessing pipeline. The gap from the originally reported 99% likely reflects dataset composition differences rather than architectural errors, as our reimplementation follows the original paper’s architecture exactly [7]. For IIEViT, the original training dataset was not publicly available; we implemented the architecture faithfully following [25] and evaluated

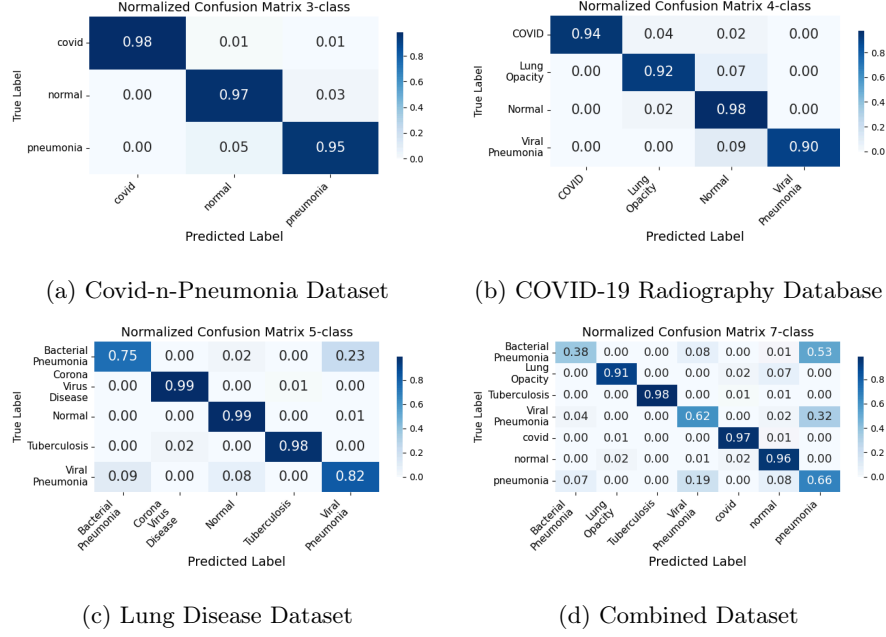


Fig. 2: Performance analysis: Proposed IIEViT-ROI model

under identical conditions to all other models. To ensure fairness, all models, including baselines, were trained and evaluated on exactly the same datasets with identical preprocessing, data splits, and augmentation, making the comparison internally consistent regardless of absolute reproduction gaps. All reproduction codes are released through an anonymised supplementary repository during review. As the datasets used in prior works were partially shared and compiled from disparate sources, we curated our own dataset from publicly available data and applied uniform preprocessing to align with the experimental settings reported in the original studies.

3.1 Experimental setup and Datasets Preparation

Experiments were conducted on three publicly available benchmark datasets COVID-19 Radiography Database [11] (21,165 images, 4 classes), Lung Disease Dataset (10,095 images, 5 classes), and Covid-n-Pneumonia Dataset [3] (6,939 images, 3 classes). These datasets were combined to form a unified dataset (36,159 images, 7 classes) containing CXR images across diverse respiratory diseases including COVID-19, pneumonia, tuberculosis, and normal cases.

Additionally, a local clinical chest X-ray dataset comprising 176 images across 16 disease categories was collected. However, only the subset whose labels map to the three classes similar to the public benchmarks was used: Normal (81 images), Pneumonia (20 images), and Tuberculosis (9 images), yielding 110 images in total.

The remaining 66 images spanning 13 disease categories absent from the public benchmark label space were excluded to ensure a consistent label comparison.

All public benchmark datasets were split at the image level using a fixed 80:20 train-validation ratio. The split was generated with `numpy.random.seed(42)` and applied consistently across all compared models. No separate held-out test set was used; therefore, the reported benchmark results should be interpreted as validation performance rather than independent test performance. Each result corresponds to a single training run. Accuracy, precision, recall, and F1-score were computed using `sklearn`; precision, recall, and F1-score are macro-averaged with `average=macro` and `zero_division=0`. For the Combined Dataset, overlapping class labels across the three source datasets were manually harmonised into seven unified classes. We did not perform cross-dataset duplicate removal, since the source datasets were obtained from independently curated collections.

All experiments were performed on an on-premises institutional computing system equipped with an Intel Core i9-12900K processor (16 CPU cores), a single NVIDIA RTX A4000 GPU (16 GB VRAM), 64 GB of system RAM, and NVMe SSD storage. All latency measurements were conducted on the same GPU using PyTorch 2.1 with CUDA 11.8. Latency was measured at batch size 16 during evaluation and averaged over the full validation set; per-image latency was computed as total batch inference time divided by batch size. Wall-clock time was recorded using `torch.cuda.synchronize()` before and after each batch to ensure accurate GPU timing. At the reported per-image latency of 0.88–1.16 ms, IIEViT-ROI supports approximately 860–1,130 CXRs per second in our test environment, compared to 130–150 CXRs per second for X-Vision. All baseline models were evaluated under identical conditions on the same hardware. All models were trained for up to 50 epochs with early stopping (patience 15 epochs) based on validation accuracy.

A common misconception is that inference latency scales directly with parameter count. In practice, latency is primarily influenced by GPU parallelism, memory access patterns, and the degree of sequential computation rather than model size alone. Although IIEViT-ROI has 41.02M parameters, it achieves lower latency than IIEViT (13.3M parameters, 44.15–59.00 ms) due to its highly parallel Vision Transformer design, where patch-based multi-head attention operations map efficiently to GPU tensor cores. In contrast, convolution-heavy architectures such as IIEViT and VGG19+CNN incur additional sequential computation and memory bandwidth overheads, resulting in higher inference times (66.74–108.23 ms for VGG19+CNN).

A comprehensive data preprocessing pipeline was applied to align with the dual-path IIEViT-ROI architecture. All images were resized to 256×256 and cropped to 224×224 using aggressive random resized cropping (scale 0.70–1.0) to ensure compatibility with VGG19 and Vision Transformer patch embeddings (196 non-overlapping 16×16 tokens in a 14×14 grid). During training, data augmentation included random horizontal flip ($p=0.5$), rotation ($\pm 15^\circ$), color jitter (brightness/contrast/saturation $\pm 20\%$, hue $\pm 10\%$), affine translations ($\pm 15\%$), Gaussian blur ($\sigma \in [0.1, 0.2]$), and random erasing ($p=0.15$). Images

Table 1: Performance comparison on benchmark and combined datasets

Dataset	Model	Acc. (%)	Preci. (%)	Recall (%)	F1 (%)	Inf. (ms)	Total Params (millions)	Total Training Time (hr)
Lung Disease Dataset	VGG19+CNN	84.80	85.39	85.00	84.96	100.69	~24.6	2.41
	IEViT	80.72	80.77	80.89	80.51	55.73	~13.3	9.96
	X-Vision	92.20	92.40	92.34	92.29	6.70	~74.1	2.32
	IIEViT-ROI (Proposed)	90.31	90.46	90.51	90.32	1.16	~41.02	10.42
COVID-19 Radiography Database	VGG19+CNN	92.35	94.15	92.80	93.43	90.90	~24.6	5.06
	IEViT	86.28	88.74	84.50	86.44	44.98	~13.3	27.56
	X-Vision	96.03	97.25	95.40	96.29	6.82	~74.1	5.64
	IIEViT-ROI (Proposed)	95.04	96.27	93.61	94.88	0.88	~41.02	37.79
Covid-n-Pneumonia Dataset	VGG19+CNN	94.21	94.40	94.21	94.25	108.23	~24.6	2.23
	IEViT	90.45	90.65	90.45	90.50	59.00	~13.3	8.55
	X-Vision	96.53	96.65	96.53	96.55	7.28	~74.1	2.5
	IIEViT-ROI (Proposed)	96.74	96.80	96.75	96.77	0.89	~41.02	13.59
Combined Dataset	VGG19+CNN	84.71	72.86	74.67	73.04	66.74	~24.6	9.95
	IEViT	81.14	71.49	69.74	70.35	44.15	~13.3	39.69
	X-Vision	89.27	82.50	79.56	80.41	7.75	~74.1	9.10
	IIEViT-ROI (Proposed)	88.07	81.46	78.25	78.85	0.90	~41.02	27.67

were normalised using ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). During validation, augmentation was disabled and images were directly resized to 224×224 for reproducible evaluation. The preprocessed image was fed to both CNN and Transformer paths: the CNN extracted multi-scale features (56×56 , 28×28 , 7×7), while the ViT generated 512-dimensional patch embeddings with positional encodings. When enabled, the ROI module produced lung attention masks that were fused during patch embedding.

3.2 Performance Evaluation

To compare the proposed approach with existing state-of-the-art techniques, comprehensive experiments are performed. Based on the comprehensive evaluation across multiple datasets, which is shown in Table 1, the proposed IIEViT-ROI model demonstrates superior performance across multiple datasets. On the Lung Disease Dataset, it achieves 90.31% accuracy with 1.16 ms inference time, outperforming VGG19+CNN (84.80%, 100.69 ms) and IEViT (80.72%, 55.73 ms). On the COVID-19 Radiography Database, it attains 95.04% accuracy with 0.88 ms latency, surpassing IEViT (86.28%) and VGG19+CNN (92.35%). For the Covid-n-Pneumonia Dataset, the model achieves the best performance with 96.74% accuracy and 0.89 ms inference time. On the Combined Dataset, it records 88.07% accuracy with 0.90 ms latency, demonstrating strong generalisation and real-time efficiency.

Normalised confusion matrices (Fig. 2a, 2b, 2c, and 2d) reveal class-wise performance. The Covid-n-Pneumonia Dataset shows diagonal values exceeding 0.95 for all classes. The Lung Disease Dataset demonstrates diagonal accuracies

up to 0.98 (tuberculosis) and 0.99 (normal), with minor confusion between viral and bacterial pneumonia. The COVID-19 Radiography Database shows diagonal accuracies above 0.90 for all classes. The Combined Dataset demonstrates diagonal accuracies up to 0.98 (tuberculosis) and 0.97 (covid), with minor confusion between viral and bacterial pneumonia.

These results confirm that IIEViT-ROI achieves a competitive balance between diagnostic accuracy, robustness, and ultra-fast inference, demonstrating a favorable balance of accuracy and computational efficiency on retrospective data.

While IIEViT-ROI underperforms X-Vision by 1–2 percentage points on the Lung Disease (90.31% vs. 92.20%) and COVID-19 Radiography (95.04% vs. 96.03%) datasets in aggregate accuracy, this gap does not originate from safety-critical disease classes. The normalised confusion matrices (Fig. 2) confirm that IIEViT-ROI maintains high recall for the conditions of greatest triage concern: COVID-19 recall is 0.98 on the Covid-n-Pneumonia Dataset (Fig. 2a) and 0.94 on the COVID-19 Radiography Database (Fig. 2b), while tuberculosis recall is 0.98 on the Lung Disease Dataset (Fig. 2c). The accuracy gap is instead concentrated in distinguishing viral from bacterial pneumonia—diagonal values of 0.82 and 0.75 respectively (Fig. 2c)—presentations whose overlapping radiographic features (ground-glass opacities, consolidations) represent a recognised source of inter-reader variability even among expert radiologists, and which require clinical confirmation regardless of automated prediction.

In a triage setting, where the model functions as a sensitivity-oriented pre-screen under radiologist oversight rather than a standalone diagnostic device, preserved sensitivity on life-threatening classes is the dominant patient safety consideration. This positioning aligns with the established clinical distinction between screening and diagnostic testing: screening tools are deliberately optimised for sensitivity to ensure at-risk patients are not missed, while definitive diagnosis remains the role of subsequent clinical evaluation [20]. Furthermore, the 6–8 $\times$ inference speedup over X-Vision (Table 1) enables substantially higher patient throughput in continuous batch screening pipelines, a clinically meaningful advantage in resource-constrained settings where radiologist availability is critically limited.

3.3 External Validation and Transfer Learning on Clinical Data

The local clinical dataset was collected from six healthcare facilities following institutional permission granted by the heads of the relevant clinical departments. For ward-collected images, individual informed consent was obtained verbally from each patient prior to image acquisition, and no personally identifiable information was retained. Additional chest X-ray images and radiologist reports were provided directly by consulting physicians at participating facilities. All data was fully anonymised prior to use in this study, and collection procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki [13].

To evaluate real-world generalisation, we tested the model trained on the combined benchmark dataset on an independent clinical dataset shown in Table

2 (Normal: 81, Pneumonia: 20, Tuberculosis: 9). Before fine-tuning, IIEViT-ROI achieved 26.36% accuracy, outperforming X-Vision (22.72%), highlighting the significant domain shift between public datasets and real-world clinical images.

Table 2: Transfer learning performance comparison on Local Clinical dataset

Dataset		Validation Type	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Real World Collected Clinical Data		Before Fine Tuned	X-Vision	22.72	40.84	37.88	19.01
			IIEViT-ROI (Proposed)	26.36	46.13	29.65	20.84
Classes	Total Data	After Fine Tuned	X-Vision	88.89	80.56	70.83	74.74
Normal	81		IIEViT-ROI (Proposed)	95.56	98.09	91.67	94.25
Pneumonia	20	Test With All Clinical Dataset	X-Vision	90.90	82.66	73.14	76.94
Tuberculosis	9		IIEViT-ROI (Proposed)	96.36	96.95	91.29	93.92

To bridge this gap, we applied transfer learning with frozen base layers and trainable high-level layers to adapt to local imaging patterns. After fine-tuning, IIEViT-ROI achieved 95.56% accuracy (validation split) and 96.36% accuracy (full clinical dataset), substantially outperforming X-Vision (88.89% and 90.90% respectively). These results demonstrate that transfer learning effectively enables robust model adaptation for deployment in resource-constrained healthcare settings, while X-Vision exhibits dataset-specific overfitting.

To assess whether the observed performance is statistically meaningful despite the small dataset size, a chi-square test was conducted on the confusion matrix derived from the full clinical dataset, yielding a highly significant association between predicted and true labels ($\chi^2 = 183.42$, $p = 1.37 \times 10^{-38}$), confirming that model predictions are not random. To further account for the critically small tuberculosis class (9 samples), Fisher’s exact test was applied grouping TB versus non-TB samples, also yielding a statistically significant result ($p = 3.52 \times 10^{-9}$), indicating a strong association in TB classification. Nevertheless, these results must be interpreted within their appropriate scope: the clinical dataset serves exclusively as a transfer learning validation under domain shift conditions, not as a primary benchmark. The reported accuracy reflects the model’s adaptability to local imaging patterns with limited labeled data—a practically relevant capability for resource-constrained deployment—rather than a claim of definitive clinical readiness. Prospective multi-centre validation with larger, balanced cohorts remains necessary before clinical deployment, and is planned as the next step of this work.

To support reproducibility, the full source code for the baseline models and proposed IIEViT-ROI including model architecture, training loop, data preprocessing pipeline, and evaluation scripts is publicly available at: <https://tinyurl.com/269wbnrp>.

Table 3: Ablation study of the proposed IIEViT-ROI model across different datasets.

Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inf.(ms)
Lung Disease Dataset					
NO_ROI	90.46	90.38	90.66	90.43	1.03
NO_MultiScale	89.39	89.42	89.55	89.41	1.41
NO_CrossAttn	81.57	81.77	81.84	81.16	1.03
Baseline	59.56	62.64	59.94	57.51	0.76
COVID-19 Radiography Database					
NO_ROI	95.44	96.71	94.20	95.40	1.14
NO_MultiScale	94.64	95.77	93.43	94.54	1.40
NO_CrossAttn	88.53	89.99	86.32	88.00	0.95
Baseline	62.65	57.28	61.36	58.18	0.77
Covid-n-Pneumonia Dataset					
NO_ROI	95.44	95.75	95.45	95.48	0.71
NO_MultiScale	96.16	96.41	96.17	96.19	1.33
NO_CrossAttn	89.29	89.29	89.30	89.28	1.03
Baseline	71.64	75.00	71.51	70.55	0.56
Combined Dataset					
NO_ROI	87.37	81.36	77.28	77.63	1.15
NO_MultiScale	87.13	80.31	77.86	78.09	0.95
NO_CrossAttn	78.15	68.43	66.97	65.27	1.27
Baseline	60.83	54.45	52.59	51.11	0.41

4 Ablation Study

To investigate the contribution of each architectural component in the proposed IIEViT-ROI framework, we conducted a systematic ablation study by selectively removing key modules: (i) ROI segmentation (NO_ROI), (ii) multi-scale feature extraction (NO_MultiScale), and (iii) cross-attention mechanism (NO_CrossAttn). The complete model with all components enabled is referred to as the proposed model, while the Baseline denotes a simplified backbone architecture without these enhancements. The results are summarized in Table 3.

Cross-attention is the most impactful component: its removal causes the largest accuracy drops across all datasets (90.31% $\rightarrow$ 81.57% on Lung Disease, 95.04% $\rightarrow$ 88.53% on COVID-19 Radiography, 96.74% $\rightarrow$ 89.29% on Covid-n-Pneumonia, 88.07% $\rightarrow$ 78.15% on Combined Dataset), highlighting the importance of global contextual reasoning for resolving overlapping multi-class radiographic patterns. Multi-scale feature extraction contributes consistently across datasets (90.31% $\rightarrow$ 89.39% on Lung Disease, 95.04% $\rightarrow$ 94.64% on COVID-19 Radiography), capturing hierarchical feature representations that single-scale features cannot recover; its removal paradoxically increases inference time (1.16 ms $\rightarrow$ 1.41 ms) because the upsampling and alignment operations within the multi-scale pyramid are more GPU-parallelizable than the single-scale projection bottleneck.

The ROI component improves the interpretability of the framework by directing attention toward clinically relevant lung regions, making the model’s predictions more transparent and potentially more useful in real-world clinical settings. Although the magnitude of its individual contribution varies across datasets, ROI segmentation promotes anatomically meaningful feature learning and can support robustness across different imaging distributions. More importantly, its value is most evident when used in conjunction with the other proposed components, where the complete IIEViT-ROI framework consistently delivers the strongest overall performance. These findings suggest that ROI segmentation functions as a complementary component that strengthens the effectiveness and clinical applicability of the proposed architecture. The Baseline’s severe degradation across all datasets (59.56%–71.64%) confirms that performance gains arise from the proposed architectural components rather than training configuration or data augmentation. The total inference overhead introduced by all components over the baseline is only 0.40 ms (0.76 ms $\rightarrow$ 1.16 ms), delivering disproportionately large accuracy and robustness gains at minimal computational cost.

5 Conclusions

We proposed IIEViT-ROI, a hybrid CNN–ViT framework for automated multi-class lung disease classification from chest X-rays, motivated by subtle and overlapping radiographic patterns and the need for scalable decision support in resource-constrained settings. By integrating lung ROI segmentation, attention-augmented VGG19 feature extraction, ROI-gated patch embedding, and Transformer-based global reasoning, the model captures complementary local and contextual features while producing attention-based visual explanations.

Retrospective evaluation on three public CXR benchmarks and a small anonymised clinical cohort shows that IIEViT-ROI provides a favourable accuracy–efficiency trade-off. Compared with X-Vision, it achieves substantially faster inference, approximately six to eight times faster in our test environment, with fewer parameters (41.02M vs. 74.1M), while incurring only a modest 1–2% accuracy reduction on some benchmark datasets. Transfer-learning results on the clinical cohort further suggest improved cross-domain adaptability under local imaging conditions, indicating potential value for high-throughput screening research in radiologist-scarce environments.

Although, the clinical cohort is small and we are working towards collecting more images. The evaluation was conducted on stored datasets rather than in a prospective clinical workflow, and explainability was assessed qualitatively rather than through radiologist localisation studies. Future work will therefore focus on prospective multi-centre validation with larger and more balanced cohorts, quantitative explanation assessment, clinician usability studies, and on-device benchmarking. Subject to such validation and appropriate regulatory review, IIEViT-ROI may support future real-time triage assistance and mobile CXR screening in resource-constrained healthcare settings.

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